

1 **Journal**

2 Communications Earth & Environment

3
4 **Title:** High-resolution analysis of power plant land requirements for net zero in the Western United States
5 indicates coordinated land use policies will be essential

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11
12 **Abstract**

13 Transitioning towards a decarbonized power grid and net-zero economy will require unprecedented
14 expansion of energy infrastructure, leading to extensive development footprints and land use conversion.
15 Understanding where future power plant sitings are likely to take place and identifying the potential conflicts
16 and land-use tradeoffs that may arise will be key to identifying feasible pathways and evaluating regional
17 planning coordination needs. Through spatially explicit integrated modeling we develop projected 1 km²
18 resolution power plant sitings in the Western United States (US) for both business-as-usual and net zero
19 pathways. We find that 30% more land will be needed for solar and wind infrastructure to achieve a net
20 zero economy over business-as-usual and that 75% of that development is projected to be located within
21 10 km of natural areas. Additionally, we find California to be the most vulnerable Western US state to
22 evolving siting constraints.

1 Introduction

Most CO₂ emissions (73%) in the United States (US) come from the combustion of fossil fuels, 31% of that from the electric power sector alone.¹ Transitioning towards a decarbonized electric grid or net zero economy will require unprecedented deployment of clean technologies such as solar and wind.²⁻⁵ Given the considerable power density per unit of land differences between renewables and dispatchable generating resources, land use considerations are poised to play a key role in this transformation.⁶ A traditional utility-scale natural gas combined-cycle facility, for example, can typically be sited within a single square kilometer (km).⁷ To site the equivalent amount of solar or wind capacity can require 15-40x the land.⁸ Despite the urgency of identifying feasible net zero pathways, there has been limited quantitative analysis of the land-use implications associated with this transition.

To deploy renewables at the projected scale necessary to achieve net zero goals will require consideration beyond individual plants (such as permitting) with potential tradeoffs across a range of institutions and parties.^{2,9} As solar and wind deployment has increased in recent years, there has been a growing number of concerns regarding large renewable projects. Studies have shown that land use change associated with renewable development can impact biodiversity and the behavior of species.^{10,11} There is a variety of research that highlights the negative human and social impacts of development such as changes to the character of the landscape and job dislocation from energy transitions.¹² New energy infrastructure development can also lead to positive economic impacts such as job growth and local economic stimulation and it is important to consider all associated costs and benefits. New policies are emerging that restrict siting based on agricultural, environmental, viewshed, and other concerns. Alberta, Canada, for example, implemented a policy banning wind development within 35 km (~21 miles) of “pristine views” and is prioritizing agriculture in their decision-making for future energy expansion.¹³ Many similar policies are being considered or have already been established in the US,¹⁴⁻¹⁶ including areas that have high renewable resource potential. San Bernardino County, a region just northeast of Los Angeles, California, banned large-scale solar and wind development within an area of over a million acres (~4,000 km²) to protect desert landscapes from utility-scale infrastructure.¹⁷ These types of siting restrictions are appearing at an accelerated rate across the country, underscoring the need for more detailed and anticipatory analysis.¹⁸

Previous studies of land-use conversion and impact have modeled energy infrastructure expansion in a variety of ways. Some studies evaluate historic buildouts and trends in wind and solar farms, and then use those findings to infer land use impacts under decarbonization.⁵ The expectation set by these studies is that recent developments are strong indicators of future development barriers and siting opposition, which may or may not be the case. Other research has evaluated socioeconomic considerations, local attitudes on development, and agricultural land preservation. These studies are typically conducted through extensive survey-based approaches which can be limited in their scope of inquiry given the corresponding respondent burden.¹⁹⁻²¹ Bessette and Mills¹⁹ evaluated the likelihood of support or opposition to new wind siting in US Midwestern states based on agricultural, land use, and demographic characteristics and found

that communities with limited natural areas and production-oriented farming are correlated with lower siting opposition. Their findings signal the need to consider multiple factors in addition to natural area conservation when identifying feasible siting locations.

There is also a diverse body of research that explores required capacity expansion buildouts under future net zero or decarbonization scenarios²²⁻²⁵ and examines the trade-offs among energy, agriculture, and natural spaces to pinpoint optimal locations for new energy infrastructure. Many of these studies establish siting exclusions to determine which spaces are suitable or unsuitable for development based on common technology and policy-based constraints (e.g., avoid National Parks), but they typically do not identify the impact of new exclusions, combinations of new exclusions, or variations in their intensity.^{18,23,26-29} Renewable siting policies are changing rapidly across the US and historic considerations may no longer be reflective of future conditions. Exploring and quantifying the uncertainty regarding the available siting landscape and the related implications for the achievement of net zero generation goals will be critical. Additionally, many of the studies using multi-year projected capacity expansion plans to identify land use impacts from power plant siting assume static conditions over the modeling period such as population density, renewable resource availability, and energy market demand. Quantifying more realistic land use impacts under a net zero pathway will require modeling structures that capture dynamic grid processes and emulate “on-the-ground” infrastructure decision making while incorporating climate, policy, and socioeconomic projections.

Here we present projected infrastructure siting and land use in the Western US through 2050 on a 1 km² grid, under both a net zero economy and a business-as-usual pathway, while considering dynamic siting constraints and opportunities. We use a geospatial power plant siting model that considers the economic tradeoff between grid interconnection costs and operating value within suitable siting locations (see Methods). The projected sitings, developed using an integrated modeling framework with dynamic population, renewable resource availability, and energy market demand and with current siting exclusions, are further evaluated for their exposure to potential changes in siting exclusion policies, specifically, the exclusion of 1) federally identified disadvantaged communities (DACs), (2) important farmland areas, and (3) areas in close proximity to natural areas

Our research is motivated by three key questions. First, how does the future power plant landscape evolve using our integrated, multisector dynamics approach? Second, given siting projections of technologies under the net zero and business-as-usual scenarios, how much land used for projected new power plant development intersects with areas that may see more restrictions in the future, such as DACs, important farmland, and land near natural areas? Last, given current and potential future siting constraints, does the Western US have sufficient suitable land capable of providing the renewable generation required to achieve our net zero goals and how sensitive is the region to various combinations of new policies?

Overall, we find that over 30% more land will be needed in the Western US by 2050 to support new solar and wind infrastructure under a net zero economy compared to a business-as-usual scenario. Additionally, of the approximately 25,000 km² of land required to support all new power plant infrastructure

under the net zero scenario, nearly half of is projected to be located in DACs and three quarters is located within 10 km of a natural area. Last, our sensitivity analysis of future solar and wind generation potential under new siting restrictions reveals that the Western US would be particularly sensitive to new policies that establish restrictions on siting in proximity to natural areas. We find that if new restrictive environmental, farmland, and socioeconomic policies are established simultaneously, 2050 net zero greenhouse gas emissions goals could become unachievable in many parts of the Western US and particularly in the state of California. These results indicate that more analysis and collaboration is needed between energy, environmental, agricultural, and community groups to determine appropriate siting locations and exclusion regions that consider tradeoffs for all parties.

2 Results & Discussion

In this section we first provide an overview of our projected power plant capacity and siting locations through 2050 and show the extent of land required to support new power plant infrastructure under net zero and business-as-usual policy futures given current siting policies. Next, we identify the intersection of projected power plant locations with three types of land not typically used to exclude power plant siting: DACs, important farmland, and land in close proximity to natural areas. Last, we evaluate a range of siting exclusion policy scenarios to understand their potential impacts on new solar and wind generation and the resulting ability or inability to meet net zero generation goals in the Western US in 2050.

2.1 30% more land dedicated to solar and wind is needed in the Western US to achieve a net-zero economy compared to business-as-usual

Results from the capacity expansion analysis show that approximately 85% of new power capacity deployed in the Western US by 2050, under either a net zero or business-as-usual scenario, will consist of solar photovoltaics (PV) and onshore wind (Figure 1). Across the eleven states representing the Western US electric grid, we see over 260 GW of new solar PV and nearly 220 GW of new onshore wind capacity deployed by 2050 under the net zero pathway. To put this in perspective, the 2021 installed power capacity of *all* technologies (renewable and non-renewable) in the US Western Interconnection was 286 GW.³⁰ In contrast to net zero, the business-as-usual scenario has 30% less solar PV and onshore wind capacity across the same timeframe. Most of the capacity additions under net zero (42%) take place in California and Utah, which see nearly 150 GW and 100 GW of new additions, respectively. These projections are comparable to those seen in recent literature exploring US capacity expansion plans under similar decarbonization and net zero pathways. In Princeton University's Net Zero America study, for example, the authors project nearly 295 GW of solar additions (PV and rooftop) and nearly 250 GW of wind additions (onshore and offshore) in aggregate by 2050 for the eleven Western US states under their high electrification scenario.²²

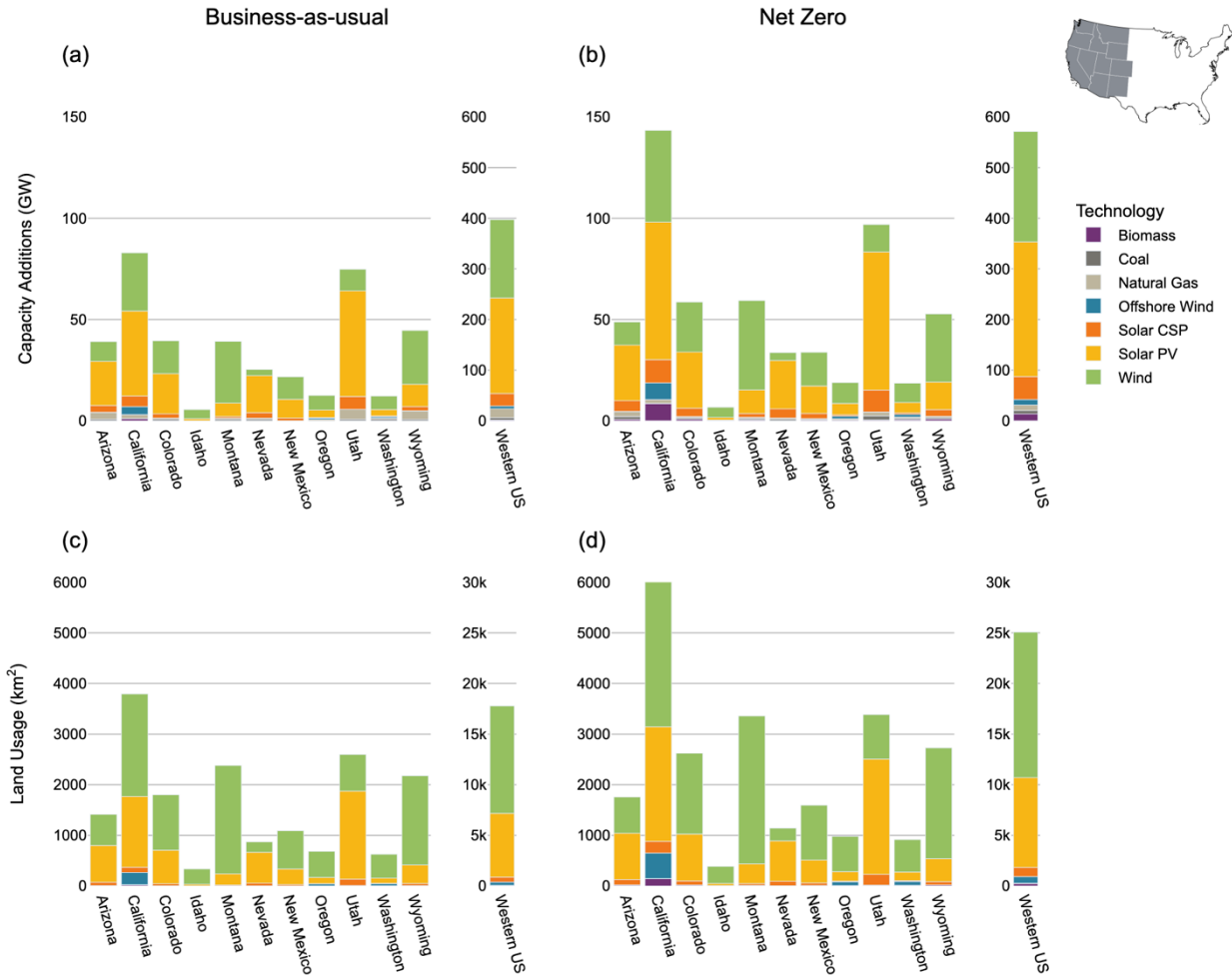


Figure 1 Cumulative new capacity additions between 2025 and 2050 by state and technology under the (a) business-as-usual scenario and (b) net zero economy scenario; and land usage (km²) required for new capacity additions through 2050 by state and fuel/technology type under the (c) business-as-usual scenario and (d) net zero economy scenario.

Comparing our scenarios in terms of total land use by 2050, the net zero scenario requires 25,002 km² across all eleven states in the Western US, whereas the business-as-usual case needs just over two-thirds of this amount, 17,736 km². For perspective, this difference (7,266 km²) is larger than US state of Delaware. At the state level, we see considerable disparities in land requirements. In California, for example, land use under the net zero scenario is more than 50% greater than in the business-as-usual case (Fig 1c and Fig 1d). Overall, California, Montana, Colorado, Wyoming, and Utah are projected to see the largest development by 2050, mostly consisting of wind and solar.

Our geospatially explicit analysis allows us to track the locations of individually sited and retired power plants under each scenario. Figure 2 presents a series of maps showing projected cumulative power plant additions and retirement locations in five-year timesteps under the net zero scenario from 2025 through 2050. A figure showing the business-as-usual case is provided in the Supplemental

146 Information (Figure SI-1). From Figure 2 we can see how clusters of solar PV and wind deployments
147 evolve over time in key subregions of each state. Notably we see large amounts of wind sited in central
148 Montana, southeastern Wyoming, southern and northern California, along the Columbia River marking
149 the border of Oregon and Washington, and in eastern Colorado. For solar PV, the larger clusters are
150 found throughout Utah, with smaller pockets scattered in California, Colorado, and Arizona.

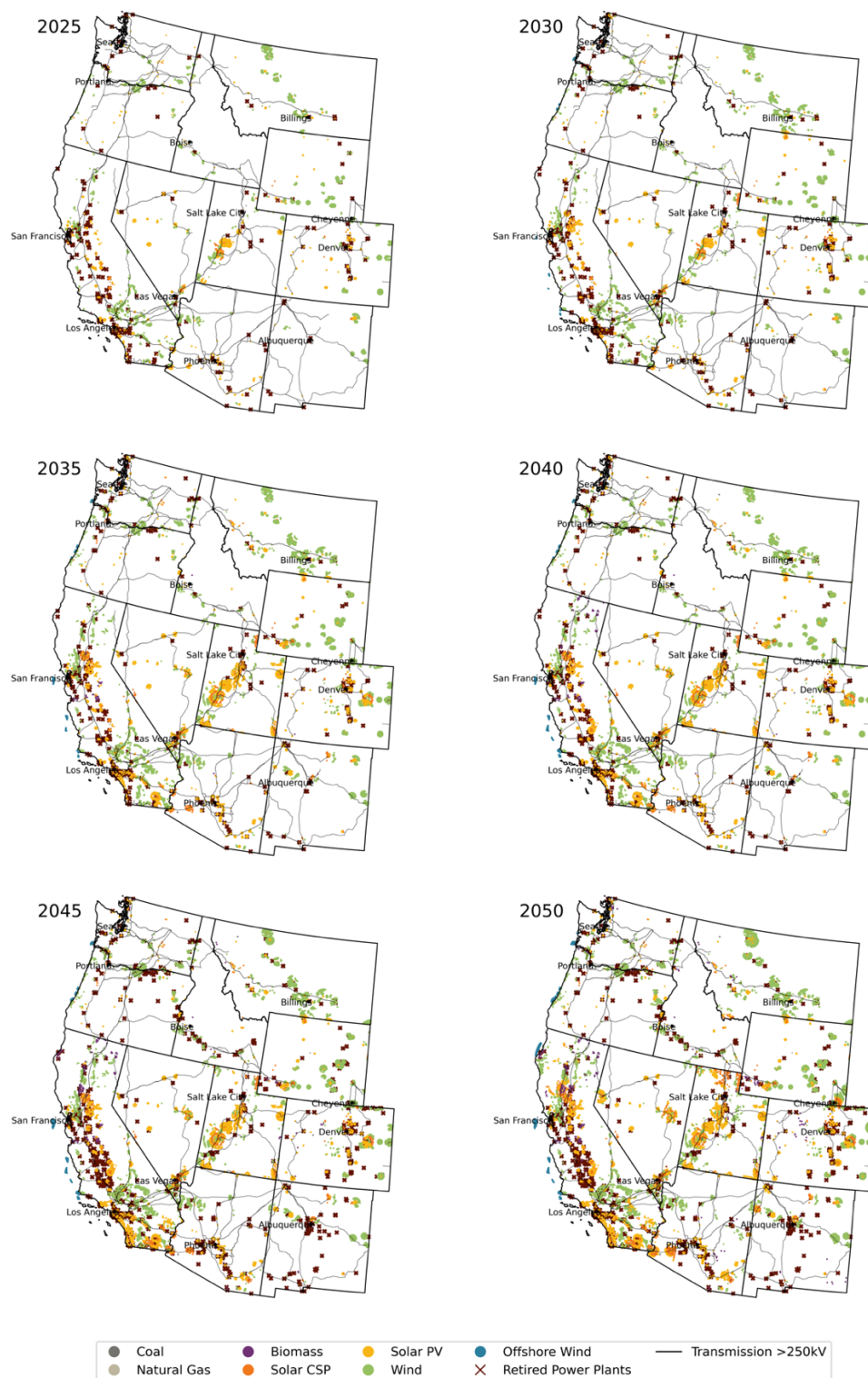


Figure 2 Map of cumulative power plant additions and retirements in the Western U.S. in 5-year time increments under the net zero scenario, 2025 through 2050. All natural gas and coal-fired generation deployed past 2035 includes carbon capture and sequestration.

2.2 More than half of all projected land use for new power plant infrastructure in the Western US by 2050 under a net zero economy is located within DACs or near natural areas

We use the geospatially explicit siting projections under the net zero and business-as-usual scenarios to evaluate how much land for new power plant development intersects with various types of land not generally used in current policies to exclude siting. The maps in Figures 3a, 3b, and 3c show cumulative power plant additions and retirements under the net zero scenario by 2050 overlaid on top of federally identified DACs (3a), combined important farmland (3b), and composite natural areas plus the surrounding 1 km area (3c). See Figure SI-2 in the Supplemental Information for the business-as-usual case.

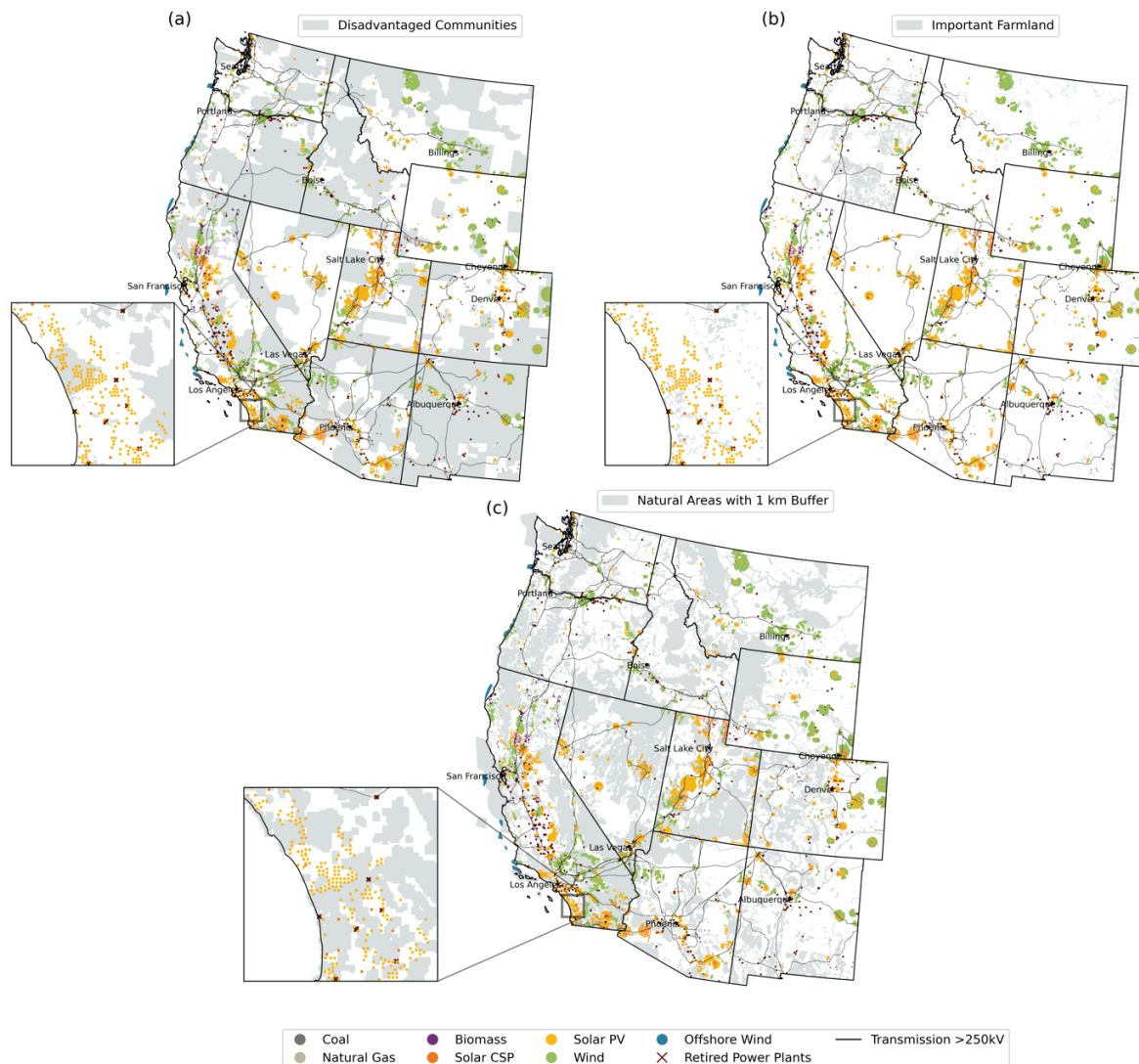


Figure 3 Map of cumulative power plant additions and retirements by 2050 in the Western US under a net zero scenario with (a) US Center for Environmental Quality-identified disadvantaged communities, (b) US Department of Agriculture-identified combined prime, state-important, locally-important, and uniquely-

important farmland, and (c) combined natural areas excluded from solar or wind siting with 1 km buffer. All natural gas and coal-fired generation deployed past 2035 includes carbon capture sequestration.

The results shown in Figure 3 have several high-level takeaways. First, we note that federally defined DACs (Figure 3a) are prominent in the Western US and can cover greater than half of state land area in some cases. An overwhelming fraction of the census tracts in the state of New Mexico, for example, are considered disadvantaged.³¹ The amount of land supporting power plant infrastructure located in DACs is projected to grow from 1,200 km² in 2020 to over 8,000 km² under business-as-usual and to over 11,800 km² under net zero by 2050. When looking specifically at solar PV and wind, we find an additional 1,783 km² of land for onshore wind and 1,690 km² for solar PV is projected to be sited in DACs by 2050 to achieve a net zero economy over the business-as-usual case (See Figure SI-3 in the Supplemental Information for all intersection values). California, Montana, and Utah especially, are projected to see increases in development within DACs for these technologies. As noted earlier, large utility-scale wind and solar development can have a range of positive and/or negative impacts at the community scale (e.g., on jobs, health, ecosystems). We find that 45% of DACs in the Western US that are projected to see the retirement of at least one coal-, oil-, or natural gas-fired power plant by 2050 also see new renewable investment (solar PV, solar CSP, or onshore wind) by 2050 under net zero (49 US census tracts in total).

Important farmland in the Western US (Figure 3b) is a composite of US Department of Agriculture-identified prime farmland, farmland of state importance, farmland of local importance, and farmland of unique importance.³² Most of the important farmland in the Western US is attributed to the category of farmland of state importance and exists mostly in Eastern Oregon, Southeastern Washington, and the California Central Valley. Under the net zero scenario, 1,689 km² for wind development and 1,089 km² for solar PV development are projected to intersect with these areas by 2050. The smaller intersection with these high value agricultural areas is a reflection that farmland qualification in the Western US is fewer than our other layers (DACs and natural layers). This result would naturally be different in the Midwest and Great Plains.

The composite natural area layer (Figure 3c) demonstrates the immensity of natural areas in the Western US. The shaded areas in this diagram represent areas that are within or less than 1 km away from land protected for preservation or conservation reasons (e.g., National Parks, National Monuments). We can see that many of the projected power plant sitings by 2050 are close to or within these areas. For example, using current siting policies, projected electricity prices, and interconnection costs, our siting model locates a large offshore wind farm within 1 km of a protected national marine sanctuary outside the San Francisco Bay Area. . Given trends in policy development and local exclusions based on natural area preservation,^{9,17,18} these types of locations may be particularly susceptible to siting opposition. Our additional analyses that looked at sitings within 5 km and 10 km of natural areas revealed a substantial amount (>4,800 km² and >5,700 km², respectively) of intersections with these areas (see Supplemental Information Figure SI-3). These analyses reveal that 75% of projected land usage for new infrastructure in

the Western US by 2050 under a net zero economy may be located within 10 km of natural areas. This indicates that natural areas are likely to be a particularly influential factor in future infrastructure siting in the region and that further evaluation of tradeoffs and alternatives may be needed to develop feasible energy transitions.

2.3 Sensitivity analysis of increasingly restrictive siting policies identifies cases where projected generation needs cannot be met, implying the need for policy coordination to meet emissions goals

When land exclusions are established, they may prevent siting in higher energy resource (e.g., high wind speed) areas. In this case, reaching the same generation target in lower resource potential locations will require a greater number of wind turbines or solar arrays, increasing both the spatial footprint of the renewable facility and the resulting land-use conversion and impact. Additionally, the exclusion of high resource potential areas makes the implicit assumption that the land remaining has adequate wind or solar resource potential to meet target generation goals—which may not be the case. We examine these issues with a series of sensitivity analysis cases.

We analyze the potential to reach net zero solar and wind generation goals in 2050 under 27 different siting restriction cases (Table 1). These cases range from our baseline siting restrictions (e.g., avoid steep slopes, avoid proximity to airports, [refer to Methods]) to cases where we add various combinations of restrictions for DACs, important farmland, three different levels of increasing environmental area protection (Level 1, Level 2, and Level 3), and three increasing amounts of spatial buffer around environmental areas (1 km, 5 km, and 10 km). Detailed descriptions of how these new restrictions were created are provided in the Methods section as well as in this manuscript's meta-repository. Tables are provided in the Supplemental Information (Tables SI-1-5) detailing the individual geospatial layers that are included in the composite exclusion areas.

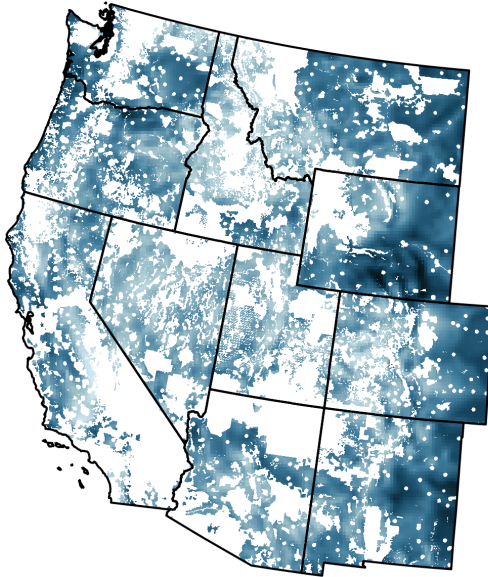
Table 1 Siting exclusion case names and corresponding excluded areas. Markings (“x”) indicate whether a particular exclusion layer is included in the given case.

#	Case Name	Technology Baseline Exclusions	Exclude DACs	Exclude Important Farmland	Environmental Area Protection Exclusions			Environmental Area Buffer Exclusions		
					Level 1	Level 2	Level 3	1 km	5 km	10 km
1	Baseline	x								
2	DAC	x	x							
3	Farmland	x		x						
4	Env_1	x			x					
5	Env_1_1km	x			x			x		
6	Env_1_5km	x			x				x	
7	Env_1_10km	x			x					x
8	Env_2	x				x				
9	Env_2_1km	x				x		x		
10	Env_2_5km	x				x			x	

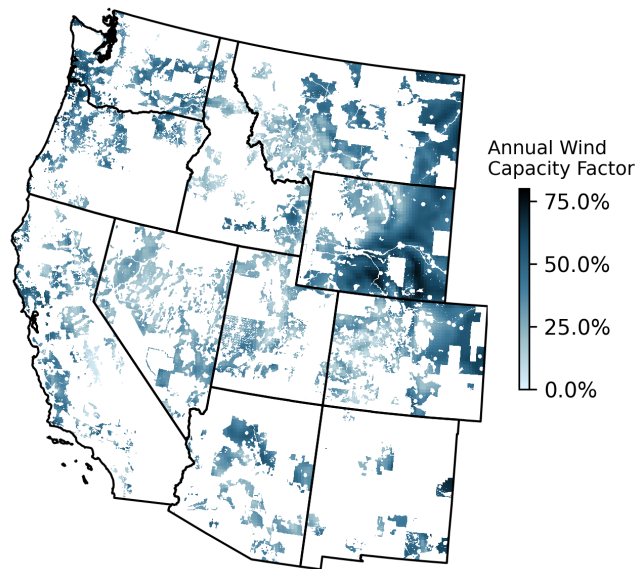
#	Case Name	Technology Baseline Exclusions	Exclude DACs	Exclude Important Farmland	Environmental Area Protection Exclusions			Environmental Area Buffer Exclusions		
					Level 1	Level 2	Level 3	1 km	5 km	10 km
11	Env_2_10km	x				x				x
12	Env_3	x					x			
13	Env_3_1km	x					x	x		
14	Env_3_5km	x					x		x	
15	Env_3_10km	x					x			x
16	DAC_Farmland_Env1	x	x	x	x					
17	DAC_Farmland_Env2	x	x	x		x				
18	DAC_Farmland_Env3	x	x	x			x			
19	DAC_Farmland_Env1_1km	x	x	x	x			x		
20	DAC_Farmland_Env2_1km	x	x	x		x		x		
21	DAC_Farmland_Env3_1km	x	x	x			x	x		
22	DAC_Farmland_Env1_5km	x	x	x	x				x	
23	DAC_Farmland_Env2_5km	x	x	x		x			x	
24	DAC_Farmland_Env3_5km	x	x	x			x		x	
25	DAC_Farmland_Env1_10km	x	x	x	x					x
26	DAC_Farmland_Env2_10km	x	x	x		x				x
27	DAC_Farmland_Env3_10km	x	x	x			x			x

As an example, Figure 4 illustrates wind and solar land availability and resource potential under our baseline case #1 and for case #19 that additionally excludes DACs, important farmland, and areas within 1 km of Level 1 environmental areas. For wind, we find that the additional siting constraints decrease available land by 39% (a 737,238 km² decrease); for solar the decrease is 27% (a 559,917 km² decrease). By comparing the projected capacity factors within the remaining suitable land, we find an overall decrease of 60% in total wind generation potential and 64% in solar generation potential in case #19 compared to the baseline siting exclusions. This shows how siting policies can severely constrain generation potential and that robust energy plans need to consider these land use-based issues.

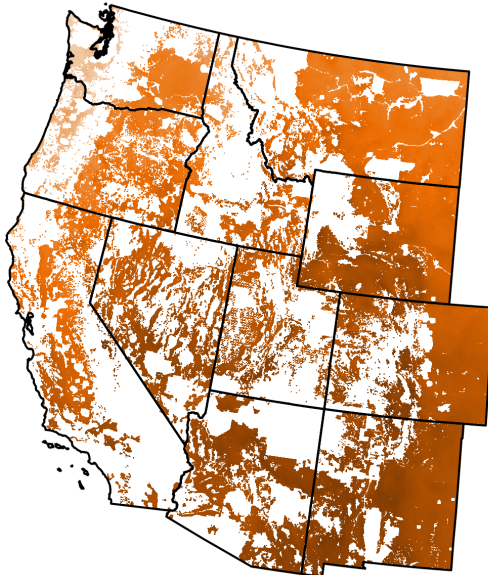
(a) Baseline Wind Siting Restrictions



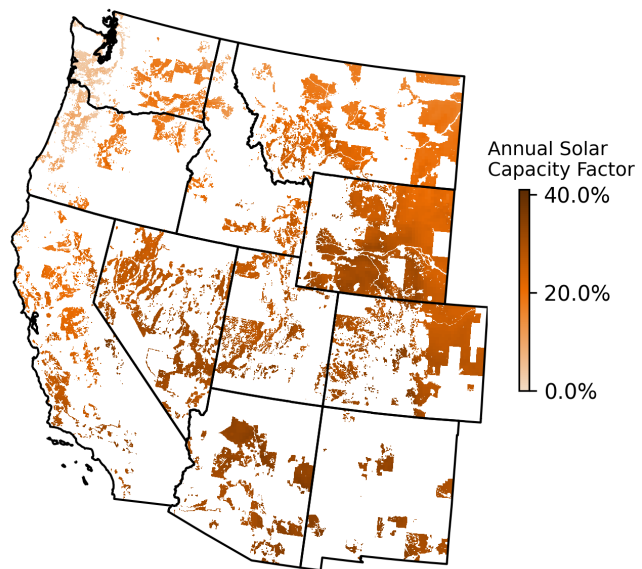
(b) Baseline Wind + Additional Siting Restrictions



(c) Baseline Solar Siting Restrictions



(d) Baseline Solar + Additional Siting Restrictions



□ Areas Excluded from Siting

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Figure 4 Annual wind capacity factor under (a) baseline wind siting suitability exclusions (case #1) and (b) baseline wind siting exclusions in addition to DACs, important farmland, and areas within 1km of natural areas; and annual solar PV capacity factor under (case #19) (c) baseline wind siting suitability exclusions

(case #1) and (d) baseline wind siting exclusions in addition to DACs, important farmland, and areas within 1 km of natural areas (case #19)

We evaluated all 27 cases described in Table 1 at the state-level, Western US level, and the electricity trading region-level (i.e., the California Grid, The Northwest Grid, and the Southwest Grid, as defined in the capacity expansion model). Results across all scales are available in the Supplemental Information (Figure SI-4 and Figure SI-5). For the remainder of our sensitivity analysis results presented here, we focus on the trading region scale because it reflects which states are able to transfer electricity given their transmission infrastructure. Figure 5 shows the projected solar and wind generation in 2050 in each grid region for each case as a multiple of the terawatt-hour (TWh) generation goal to reach a net zero economy. In cases where the available generation is to the left of the dashed vertical line (0x), we determine that there is insufficient generation potential from the remaining land available for siting to achieve net zero goals. In cases where the available generation is to the right of the dashed vertical line, enough land with adequate generation potential is available to meet net zero goals. For example, a value of 100 for wind indicates that you could generate 100x the wind generation needed to meet the net zero generation goal using the available land and its corresponding wind resource potential after all land constraints are applied in the given scenario.

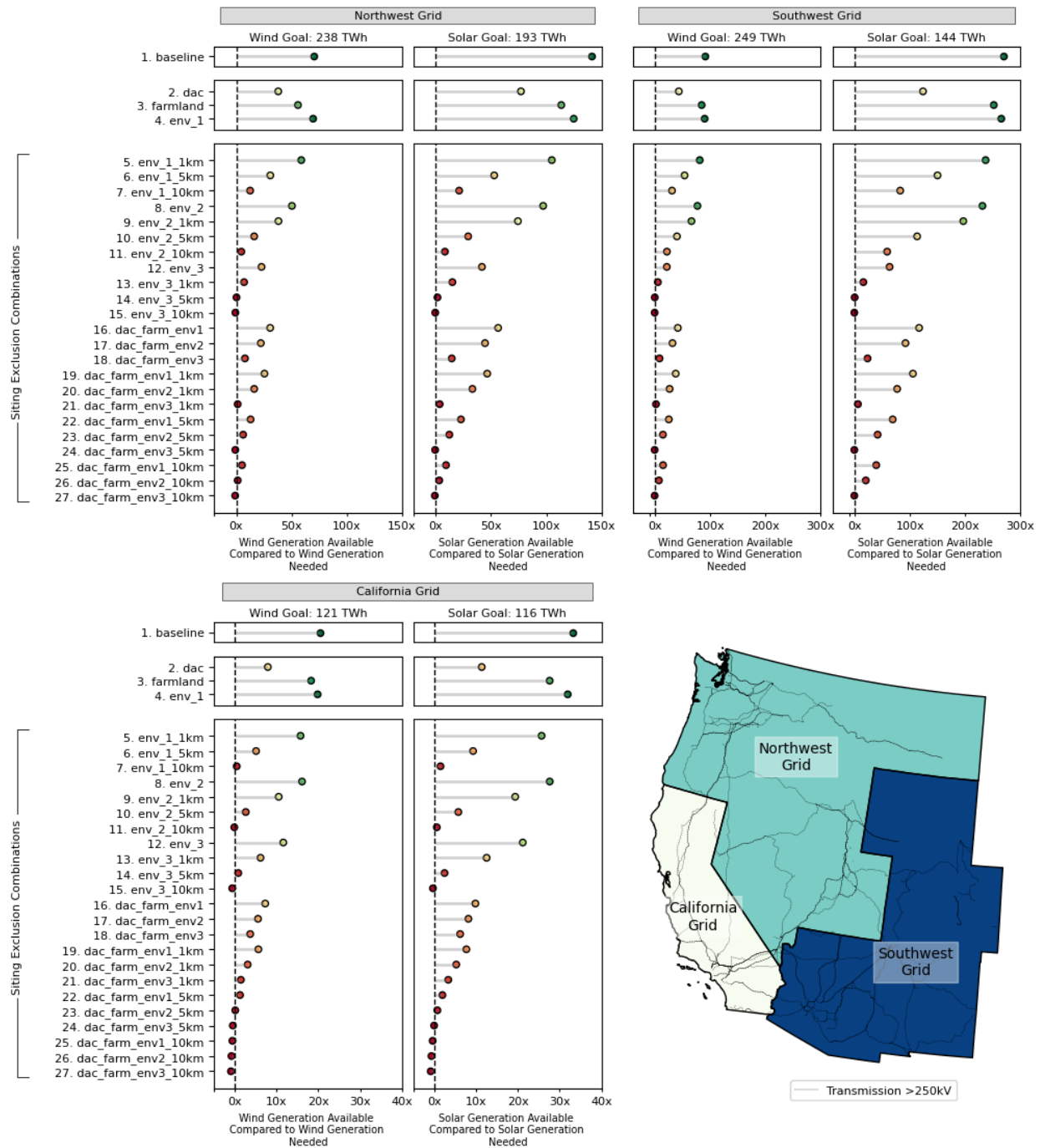


Figure 5 Available wind and solar generation (dots) compared to generation needed to achieve net zero under various siting exclusion cases by electricity trading region. Generation potential is presented as a multiplier of the generation (TWh) needed to achieve net zero goals. Cases where dots are to the left of the vertical dashed line indicate insufficient generation potential. Note that x-axis ranges differ by region. The number listed at the top of each box provides the solar or wind generation to meet the net zero generation goal in 2050 in that trading region.

Our analysis shows that in 64 out of 81 (approximately 80%) possible case-region combinations (27 cases across 3 trading regions), there is sufficient available land to meet or exceed wind generation goals for net zero by 2050. For solar generation we see sufficient land for 67 out of 81 combinations (approximately 83%) These cases generally consist of individual exclusions (e.g., farmland or DACs), combinations with lower levels of environmental exclusion (e.g., Level 1), or combinations where the environmental buffer is 1 km or less.

In the other case-region combinations (17 for wind and 14 for solar), our findings show that the remaining available land is insufficient to meet net zero generation goals in 2050. This outcome occurs most often under higher levels of environmental protection (e.g., Level 3), and when large buffers surrounding natural areas are also excluded (e.g., 10 km buffer), though the outcomes vary by trading region. The California grid region is found to be particularly sensitive to increased environmental protections and buffer extents. It is the only trading region to have insufficient land to support 2050 net zero wind generation under four scenarios: a Level 1 environment + 10 km buffer scenario (#7), a Level 2 environment + 10 km buffer scenario (#11), and the two scenarios with these restrictions plus the exclusion of important farmland and DACs (#25 and #26). The two other regions (Northwest Grid and Southwest Grid) each have sufficient wind generation under these cases. Cases in which environmental areas, DACs, and important farmland are simultaneously excluded affect land availability across all regions. For example, when important farmland, DACs, and the lowest level of environmental restriction (Level 1) with no buffer applied are all excluded (scenario #16), wind generation potential in the Northwest Grid is reduced by more than half.

The large fraction of cases with sufficient land suggests lower-level environmental, socioeconomic, and agricultural goals may be coincidentally achievable without detracting from climate objectives with adequate coordinated planning. It is worth noting that there are likely to be economic tradeoffs for renewable energy developers to avoid these areas and more analysis is needed to determine whether disincentives from an economic viability perspective would be enough to reduce total generation achievable. Larger, more expensive wind farms placed in lower capacity factor areas, for example, may be infeasible from a developmental perspective under current financial incentive structures.

Figure 5 demonstrates that wind generation is more sensitive to increasingly restrictive siting policies compared to solar PV, due in part to lower power density per unit of land assumptions. This implies a shift to solar generation to meet net zero goals may be beneficial, however, land impacts and tradeoffs from solar farms may be higher than wind farms, especially in agricultural applications as crops and solar arrays compete for sun exposure areas. Shifting onshore wind resources to offshore areas or to rooftop solar PV may be other potential alternatives, however, these are not without their own impact and cost challenges. Additional alternatives to finding land use tradeoffs for solar and wind are to deploy higher power density technologies. The net zero scenario capacity expansion projections used in this experiment do not deploy large amounts of carbon capture sequestration or nuclear technologies. These technologies

have their own costs and siting conflicts and can raise objections for several reasons in addition to the ones discussed in this paper and would not necessarily be an easier pathway to net zero.

Excluding Level 3 (moderate risk and higher) environmental areas in combination with important farmland and DACs is impactful in all three trading regions. To meet net zero goals, viewsheds, habitats, and migration corridors may be impacted. If new transmission corridors are developed between trading regions with sufficient resource availability, multiple goals may be attainable. The Northwest Grid, for example, has more wind resource availability compared to other regions under Level 3 environmental restrictions (see Supplemental Information Figure SI-4) and could export excess wind generation to the California Grid region. Transmission expansion on this scale can be complex and costly, however. For example, the Pacific Intertie consists of a set of large capacity transmission lines stretching from the northern Oregon border all the way to Southern California. The project, which came into service in the 1960s, took decades to come to fruition and cost \$700 million (equivalent to \$7 billion in 2024 USD) for the initial build, suggesting that this would not necessarily be an expeditious or low cost solution.³³

To further explore our results, we conducted a sensitivity analysis around the assumed power density of solar and wind installations. Figures are provided in the Supplemental Information (see Figures SI-6 and SI-7). When the power density of wind is reduced by half, reflecting scenarios in which local regulations may limit turbine size or when greater spacing between turbines is required, seven additional instances across grid regions and cases lead to inadequate generation to achieve net zero goals. For example, excluding DACs, important farmland, and areas within even 1 km of Level 2 environmental areas puts net zero generation goals out of reach in the California grid region. In contrast, if power density of wind were to *increase* by 50%, net zero wind generation goals become achievable for just two additional cases in the California grid. Similar results were found for solar generation under a +/- 50% power density analysis. Overall, we find that increasing power density leads to a small number of additional successful sensitivity cases, but decreasing power density leads to many more cases where we cannot achieve our clean grid goals. This implies that pathways that focus on technological improvement and power density increases can only go so far in helping to address land use conflicts.

3 Conclusions

This study presents projected power plant sitings and land use in the Western US under business-as-usual and net zero economy scenarios developed using an integrated multi-model framework that translates state-level capacity expansion plans to on-the-ground power plant sitings at a 1 km² resolution. The framework incorporates socioeconomic change and future climate impacts on electricity demand and renewable resource availability. Our analysis projects that over 7,000 km² of additional land to support new energy infrastructure will be required in the Western US by 2050 to achieve a net zero economy over a business-as-usual scenario due to extensive solar and wind buildout. We find that understanding the feasibility of such land use change may be especially critical in the states of California, Montana, Colorado, Wyoming, and Utah which are projected to see the largest wind and solar development. Furthermore, our

analysis shows that many new power plant sitings are projected to coincide with DACs or be sited near natural areas. We also find that certain combinations of new siting restrictions can greatly impact the ability to achieve net zero economy targets. Under a sensitivity case where moderate to high-risk natural spaces and the surrounding 5 km area are excluded from siting in addition to our baseline restrictions, we find that projected net zero wind and solar generation goals are unachievable in much of the Western US.

Protection of our natural areas and habitats, consideration of the wants and needs of DACs, preservation of land with agriculture potential, and decarbonizing our electric grid and economy are all important goals. Given land availability and energy resource constraints, however, it may not be possible to achieve all these simultaneously if each target is pursued independently and without consideration of their interactions and impacts. To meet future electricity needs, energy planning and analysis that considers impacts and tradeoffs from non-energy perspectives will be key to identifying feasible and robust energy futures.

4 Methods

4.1 Power plant landscape evolution under net zero and business-as-usual

This study uses results from an internally consistent, multi-model integrated framework developed by the Grid Operations Decarbonization Environmental and Energy Equity Platform (GODEEEP) project³⁴ conducted at Pacific Northwest National Laboratory. The framework downscales state-level capacity expansion projections to higher resolutions by linking the US version of the human-Earth systems Global Change Analysis Model (GCAM),³⁵ an hourly electricity demand model (Total Electricity Loads [TELL]³⁶), a geospatial power plant siting model (Capacity Expansion Regional Feasibility [CERF]^{37,38}), an hourly solar and wind generation model (Renewable Energy Potential [reV])³⁹, and an hourly grid operations model (GridView)⁴⁰. Through this multi-scale, temporally dynamic process, we project high-resolution power plant deployment and grid operations through 2050 in five-year timesteps. Figure 6 presents the multi-model framework for the experiment for a single time-step.

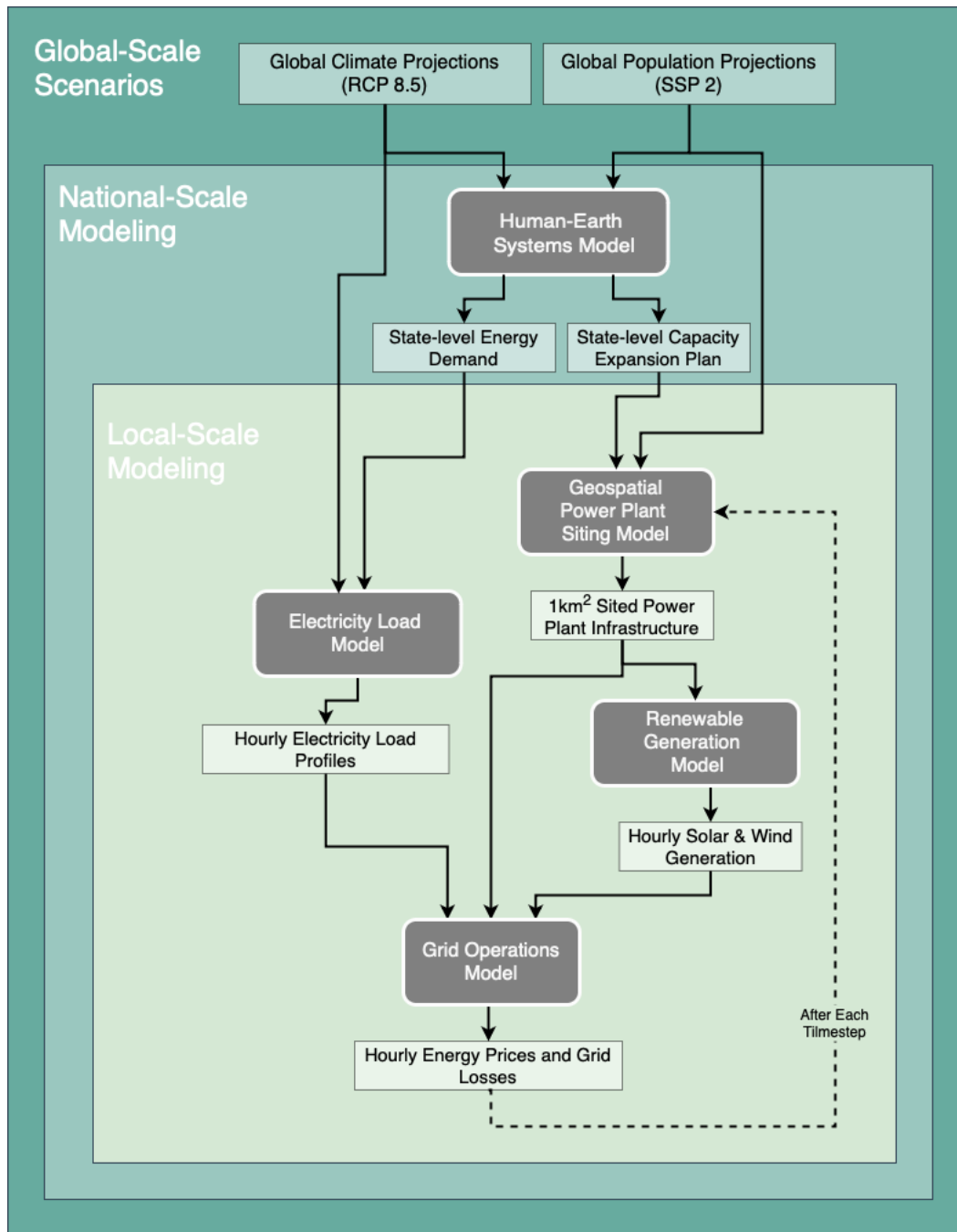


Figure 6 The Grid Operations Decarbonization Environmental and Energy Equity Platform (GODEEEP) multi-model framework.

GCAM-USA provides pathway capacity expansion plans in 5-year increments between 2025 and 2050 for both net zero and business-as-usual scenarios for 11 states in the Western US: Arizona, California, Colorado, Idaho, Montana, Nevada, New Mexico, Oregon, Utah, Washington, and Wyoming. These state-level capacity expansion plans are downscaled by CERF, which sites individual power plants on a 1 km² basis in each timestep. New renewable energy infrastructure from CERF is used in the reV model to

produce location and year-specific hourly solar and wind generation profiles. Power plant location and renewable generation profiles are handed to the nodal network grid operations model (GridView), which operates new and existing infrastructure on an hourly basis to meet hourly energy demand projections from the TELL model. GridView produces locational marginal energy prices (LMPs) based on power plant operations at each node of the Western US electric grid. These LMPs are used by CERF as part of the process to determine power plant sitings for the next timestep in the sequence. Model inputs are harmonized using a dynamically downscaled meteorological dataset following Representative Concentration Pathway (RCP) 8.5,⁴¹ and downscaled US population⁴² projections following the Shared Socioeconomic Pathway (SSP) 2 scenario for each given year of analysis. The subsections below provide more details on each of the models in the integrated chain.

4.1.1 Net zero and business-as-usual pathways (GCAM-USA)

GCAM-USA, a version of GCAM with state-level detail for the US, is an open-source, dynamic recursive, integrated assessment model that represents energy, economy, agriculture, land use, water, and climate systems.³⁵ GCAM-USA solves the equilibrium prices and quantities for over 1,000 markets (i.e., to meet different types of demands within the water, land, and energy sectors) in each state and modeling period (with 5-year time-steps). The GCAM-USA scenarios⁴ use version 6.0 of the model and encompass both business-as-usual trajectories and policy scenarios aiming for economy-wide net-zero greenhouse gas emissions by 2050. All net zero policy trajectories also entail achieving a decarbonized electricity system by 2035. These scenarios represent varying combinations of assumptions regarding climate impacts, decarbonization policies, and incentives for decarbonization. GCAM-USA analyzes a wide range of economic and sustainability indicators at the state level. Emission scenarios are based on the energy and climate targets for the US, considering different assumptions about the availability of carbon capture and storage technologies, as well as other methods such as direct air capture and measures to reduce energy demand through efficiency improvements and behavioral changes.

4.1.2 Power plant siting model (CERF)

CERF is a high-resolution, open-source geospatial power plant siting model³⁷ that downscales regional capacity expansion plans to evaluate the evolution of the power plant landscape over time. The model identifies feasible, location-specific sites for both renewable and non-renewable technologies using a suite of geospatial siting suitability data for dozens of specific power plant technology configurations (e.g., cooling types, turbine types, with or without carbon capture). By integrating the detailed geospatial suitability data³⁸ with an economic algorithm, CERF selects optimal plant locations based on factors like grid interconnection costs and the locational marginal value of new generation (Figure 7). These technology-specific sitings are provided on a 1 km² basis in the form of x/y coordinates using the Albers Equal Area Conic (ESRI:102003) projection. . The economic algorithm in CERF determines individual power plant locations through a competitive process, considering technology-specific costs for connecting

to the nearest substation and natural gas pipeline (if necessary), along with the technology-specific value of new generation in that location based on wholesale electricity prices from the grid operations model. Considering dynamic factors such as protected lands, population density, existing infrastructure, and water availability, CERF serves as a form of “ground-truthing” to both ensure the viability of broader expansion planning models and depict the evolution of the power plant landscape under various climate, socioeconomic, technological, and policy scenarios. Population-density based siting restrictions are consistent with SSP2 population projections in each run year.⁴²

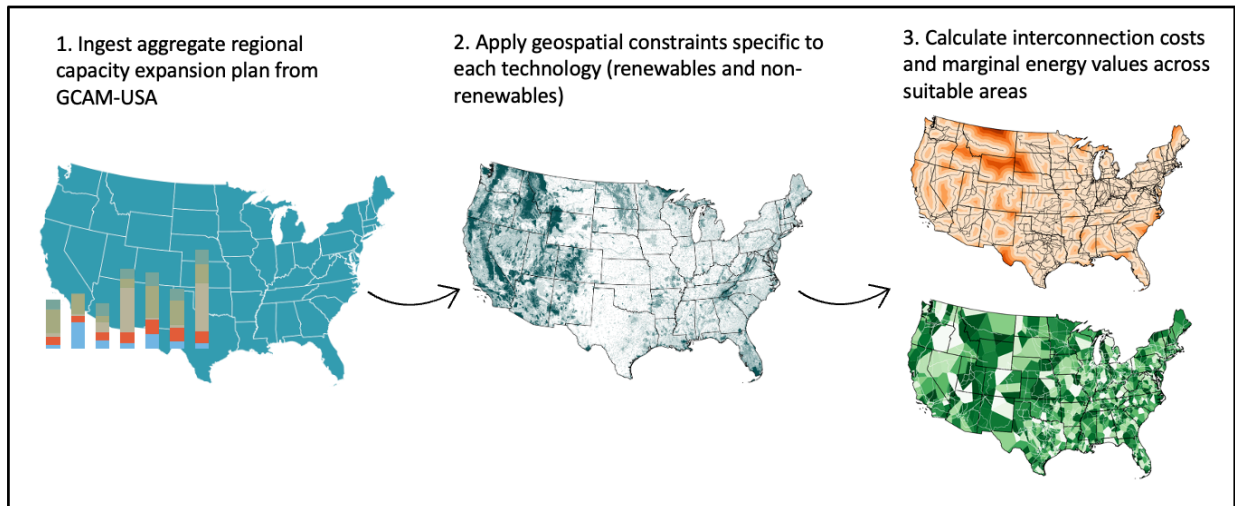


Figure 7. Capacity Expansion Regional Feasibility (CERF) model components.

4.1.3 reV renewable generation model

reV is an open-source renewable generation model that assesses resource potential given specific wind and solar farm characteristics.³⁹ In each run year, solar and wind farm configuration and location data is collected from CERF and run through reV to provide hourly resource profiles. The wind and solar models require several meteorological inputs. The wind model requires pressure, temperature, wind speed, and wind direction at the turbine hub height. The solar model requires temperature, pressure, wind speed and solar irradiance. The solar irradiance must be supplied in the form of global horizontal, diffuse normal, and diffuse horizontal irradiance. A complete description of the wind and solar data preprocessing and their evaluation is described in Campbell, et al.⁴³ Climate data used in reV to create the generation output is consistent with the RCP 8.5 (hotter) climate scenario used throughout the GODEEEP experiment.⁴¹

4.1.4 TELL hourly electricity load model

The TELL model³⁶ projects hourly electricity demand over time in response to weather and climate variability. The model is machine learning-based and was trained on historical hourly meteorology and energy demand data for 54 balancing authorities in the US. TELL takes as input the annual state-level total load projections from GCAM-USA and downscales them to hourly time-series at the county-

state-, and BA-scale. Loads across all scales coming out of TELL are conceptually and quantitatively consistent. The electricity demand projection used in the GODEEEP experiment are publicly available.⁴⁴

4.1.5 GridView

GridView⁴⁰ is a proprietary grid operations tool that models transmission and optimization of electric grid resources. The model determines appropriate utilization of generators and transmission lines based on the production cost of generation. In the multi-model framework, GridView operates existing and new generation facilities from CERF to meet hourly load projections from TELL while using renewable generation profiles from the reV model. The integrated engineering and economic analysis conducted by the model produces LMPs for each substation in its nodal network. These LMPs are used by the CERF model in subsequent timesteps to determine net operating value from each interconnection point on the grid. Note that the substation infrastructure network is assumed to remain consistent throughout the experiment timeline. GridView was chosen as the appropriate grid operations model for the GODEEEP experiment because it is a commonly used tool by long-term and operational planners in utilities across the Western US.

4.2 Evaluation of power plant siting interactions with DACs, important farmland, and natural areas

To gain insight into the amount of project power plant sitings and associated land usage that intersects with areas of interest, we use high resolution geospatial raster data layers. These areas include federally identified DACs, combined important farmland, and composite natural areas. This section describes the methodology behind the projected infrastructure siting intersection analysis. Documented code to reproduce this analysis is provided in this article's meta-repository (see Code Availability section).

First, we use CERF power plant siting projections for both the net zero and business-as-usual scenarios from the experiment workflow described in Section 4.1 to determine the coordinates of new infrastructure sitings. Second, we overlay vector point CERF sitings on 1 km² resolution geospatial raster representations of our areas of interest. Last, we use geospatial analysis to sum the number of intersections between power plant points and grid cell representations of indicated areas to determine the total number of km² of sitings by technology.

Spatial representations of DACs, natural areas, and important farmland were collected from the Geospatial Raster Input Data for Capacity Expansion Regional Feasibility (GRIDCERF) database^{38,45} which provides a binary representation of the contiguous US (CONUS) that meet or do not meet a specific criterion at a 1 km² resolution. Federally identified DACs include any census tract that is designated by the US Center for Environmental Quality (CEQ) as disadvantaged due to water, energy, climate, workforce, pollution, health, or transportation reasons.³¹ Important farmland includes any area designated by the U.S. Department of Agriculture as prime farmland, farmland of state importance, farmland of local importance, or farmland of unique importance.³² These areas have higher soil quality and moisture supply, indicating that they are potentially valuable for agricultural purposes. Natural areas are comprised of 28 unique

environmental areas from the GRIDCERF database associated with undisturbed natural areas, protected natural areas, significant viewsheds, protected habitats, and similar. These layers can include areas such as US National Parks, national marine sanctuaries, scenic and recreational areas, US Bureau of Land Management outstanding natural areas, wilderness areas, wild and scenic rivers, and many others. For the full list of layers included in the compiled natural area designation, refer to the Supplemental Information (Table SI-5).

4.3 Net zero solar PV and wind generation goal feasibility under various siting constraint combinations

We analyzed 27 different land suitability scenarios (Table 1) as part of our analysis to determine the feasibility of a net zero capacity expansion plan under additional power plant siting restrictions. We began with the baseline siting exclusions for each technology (see Supplemental Information Table SI-1 and Table SI-2) and layered additional exclusions including DAC areas, important farmland, three different levels of environment protection (Level 1, Level 2, and Level 3), three different levels of spatial buffer around environmental areas (1 km, 5 km, and 10 km), and various combinations of each.

New geospatial raster layers were created for each of the siting suitability constraint scenarios where additional restrictions were layered onto existing siting restrictions for the given technology. For example, under the DAC scenario, the composite raster layer excludes all areas from siting that are either (1) included in baseline siting exclusion areas or (2) in a federally identified DAC. The baseline siting considerations for this analysis are consistent with SSP2 populations for 2050, to capture downscaled population dynamics within the 2050 GCAM-USA simulations. Baseline siting layers for solar and wind are consistent with those identified in Mongird, et al.⁴⁵ and represent current technology and policy siting constraints.

Varying the degree of environmental area protection (Levels 1-3) is intended to determine the degree of generation potential if new habitats and natural areas are excluded from development above and beyond what is protected today. These include applying any current natural area exclusions that apply to either solar or wind to *both* solar and wind (Level 1), adding additional areas of high environmental risk (e.g., golden eagle habitats) (Level 2), and adding areas of moderate risk (e.g., golden eagle distribution) and above (Level 3). The designation of high risk and moderate risk areas is leveraged directly from U.S. Bureau of Land Management's wind development assessment of the Western US.⁴⁶ Though some of the restrictions in Level 1 only apply to solar *or* wind, we decided to include Level 1 as an analysis to capture examples of new restrictions that exclude entire subregions from any renewable development,¹⁷ regardless of whether one technology may be less impactful than another. A table with each of the individual areas included in Level 1, Level 2, and Level 3 is provided in the Supplemental Information (Table SI-5).

In addition to varying the level of natural area protection, we also tested how sensitive results are to increasing the spatial area of existing and new environmental exclusion areas. The intention of this analysis is to determine if new policies that place additional buffers on environmental areas are particularly impactful to generation goals. 1 km, 5 km, and 10 km were chosen as buffers to represent small,

intermediate, and large buffered areas. Though there are examples of new policies that stretch beyond 10 km,¹³ our results indicate that buffers beyond 10 km yield similar outcomes given the quantity of natural areas in the Western US.

To determine the resource potential within the remaining land area after each exclusion, we used 2050 capacity factor projections for solar and wind coincident with an RCP 8.5 (hotter) scenario.⁴⁷ The dataset employs regional atmospheric climate model results to produce wind and solar power generation throughout the CONUS at a 1/8th degree (12 km²) resolution which was downscaled to 1 km² for this analysis. For wind, these capacity factors are provided at the 125-meter height to be as consistent as possible with our 120-meter turbine hub height assumption. Total annual generation (in megawatt-hours [MWh]), for each region subject to siting restrictions is determined by summing the product of the assumed rated technology power capacity in megawatts (MW) and the sum of hourly capacity factors in each grid cell across all hours of the year, across all available grid cells within the region.

We assume a power density of 12 MW/km² for a 120m hub-height onshore wind technology and 30 MW/km² for solar PV. Total area includes land used directly by each technology as well as land for clearance, buildings, roads, and other areas included in the site boundary⁴⁸. A sensitivity analysis was conducted with $\pm 50\%$ power density for each technology (6 MW and 18 MW for wind, 15 MW and 45 MW for solar PV) to ascertain how results fluctuate under varying power density assumptions. The sensitivity analysis results can be found in the Supplemental Information (Figure SI-6 and SI-7) and in the paper's meta-repository.

Total generation potential was aggregated to various spatial boundaries including GCAM-USA trading region level, each of the 11 Western US states individually, and the Western US region as a whole (sum of all 11 states). Note that GCAM-USA electricity trading regions are approximated to state boundaries. There are three electricity trading regions in the Western Interconnection according to GCAM-USA: the Northwest Grid (Oregon, Washington, Idaho, Montana, Nevada, Utah), the Southwest Grid (Arizona, New Mexico, Colorado, and Wyoming), and the California Grid (California). GCAM-USA trading regions are defined within the GCAM-USA model as clusters of states that have the capability to trade energy amongst one another and follow the US Department of Energy's Energy Information Administration National Energy Modeling System Electricity Market Module Regions⁴⁹.

The target solar and wind generation for 2050 under the net zero grid in each trading region is determined by summing generation across all states within each trading region in the GODEEPP GCAM-USA simulation output.⁴ Total generation potential for solar and wind under each siting suitability scenario were compared to the target generation level to determine the ability to meet decarbonization goals. Scenarios in which available generation in a specified area was less than target net zero generation for the same specified area was an indication that the remaining available land did not contain adequate resource potential to meet generation goals in 2050.

540 **Acknowledgements**

541 This research was supported by the Grid Operations, Decarbonization, Environmental and Energy Equity
542 Platform Investment, under the Laboratory Directed Research and Development Program at PNNL. This
543 research leverages and enhances the capabilities of GCAM-USA, CERF and TELL developed under the
544 U.S. Department of Energy, Office of Science, as part of research in MultiSector Dynamics, Earth and
545 Environmental Systems Modeling Program. PNNL is a multi-program national laboratory operated for the
546 U.S. Department of Energy by Battelle Memorial Institute under Contract No. DE-AC05-76RL01830.

547 **Competing Interests**

548 The authors declare no competing interests.

549 **Data Availability**

550 All data associated with the analyses provided in this paper is publicly available for download at
551 <https://doi.org/10.5281/zenodo.13124640>.

552 **Code Availability**

553 All code used to reproduce the analysis presented in this article is openly available at the paper's
554 associated meta-repository on GitHub: https://github.com/GODEEEP/mongird-etal_2024_tbd. The meta-
555 repository includes detailed information on how to retrieve all associated input data, replicate the analysis
556 presented here, and create data visualizations shown in the manuscript.

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